PROJECT REPORT

AI Job Market Analysis Dashboard using Microsoft Excel

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1. Introduction

Project Overview

The AI Job Market Analysis Dashboard is an interactive Microsoft Excel project developed to analyze trends within the artificial intelligence job market. The project demonstrates the complete data analysis process, from raw data preparation to dashboard development, enabling users to explore salary trends, hiring patterns, experience requirements, work modes, industries, and company locations through interactive visualizations.

The dashboard allows users to dynamically filter the data using slicers, making it easier to explore different perspectives of the dataset and derive meaningful business insights.

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2. Project Objective

The objectives of this project are:

- Perform data cleaning and validation.
- Conduct data profiling.
- Develop business KPIs.
- Analyze AI job market trends.
- Build interactive Pivot Tables and Pivot Charts.
- Create an interactive Excel Dashboard.
- Generate actionable business insights.

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3. Dataset Information

Attribute	Value
Dataset	AI Job Market Dataset
Total Records	15,000
Total Columns	19
Companies	16
Job Titles	20
Industries	15
Tool Used	Microsoft Excel

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4. Data Cleaning

The following data cleaning operations were performed:

- Imported the raw dataset into Microsoft Excel.
- Converted the dataset into an Excel Table.
- Assigned a structured table name.
- Verified column headers.
- Checked for missing values.
- Checked duplicate records.
- Validated salary and experience columns.
- Ensured consistent data formatting.

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5. Data Profiling

The dataset was profiled to understand its structure and quality.

Metric	Result
Total Records	15,000
Total Columns	19
Missing Values	0
Duplicate Records	0
Companies	16
Job Titles	20
Industries	15
Average Salary	US$115,349
Highest Salary	US$399,095
Lowest Salary	US$32,519
Average Experience	6.25 Years

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6. Key Performance Indicators (KPIs)

The dashboard includes the following KPIs:

KPI	Value
Total Jobs	15,000
Average Salary	US$115,349
Highest Salary	US$399,095
Lowest Salary	US$32,519

KPI	Value
Companies Hiring	16
Industries	15
Average Experience	6.25 Years
Remote Jobs	4,920
Hybrid Jobs	5,005
On-site Jobs	5,075

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7. Pivot Tables

The following Pivot Tables were created to support analysis:

1. Average Salary by Job Title
2. AI Job Postings by Industry
3. Average Salary by Experience Level
4. AI Job Postings by Company Location
5. Work Mode Distribution
6. Top Companies Hiring AI Professionals

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8. Dashboard Visualizations

The dashboard consists of the following visualizations:

- Horizontal Bar Chart – Top Highest Paying AI Job Roles
- Column/Treemap Chart – AI Job Postings by Industry
- Line Chart – Average Salary by Experience Level
- Doughnut Chart – Work Mode Distribution
- Bar Chart – AI Job Postings by Company Location
- Horizontal Bar Chart – Top Companies Hiring

Interactive slicers allow users to filter the dashboard by:

- Industry
- Experience Level
- Work Mode
- Company Location

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9. Dashboard Features

The dashboard includes:

- Interactive KPI Cards
- Pivot Tables
- Pivot Charts
- Dynamic Slicers
- Automated Filtering
- Interactive Data Exploration
- Professional Dashboard Layout
- Business-Oriented Visualizations

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10. Key Insights

The analysis revealed several important insights:

- The dataset contains 15,000 AI job postings across 16 companies and 15 industries.
- The average salary for AI-related roles is approximately US$115,349.
- Higher experience requirements generally correspond to higher salaries.
- Remote, Hybrid, and On-site opportunities are distributed relatively evenly across the dataset.
- AI hiring is concentrated within a limited number of industries.
- Salary levels vary considerably across different AI job titles, highlighting the value of specialized roles.
- Geographic distribution of job postings indicates that AI hiring demand differs by company location.

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11. Skills Demonstrated

This project demonstrates practical knowledge of:

Microsoft Excel

- Excel Tables
- Pivot Tables
- Pivot Charts
- Slicers
- Data Cleaning
- Data Profiling
- Dashboard Development
- KPI Design
- Data Visualization
- Business Intelligence
- Exploratory Data Analysis

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12. Conclusion

The AI Job Market Analysis Dashboard demonstrates the use of Microsoft Excel for end-to-end data analysis. Starting with raw data, the project includes data cleaning, profiling, KPI creation, Pivot Table analysis, and dashboard development. Interactive slicers and visualizations enable users to explore hiring trends, salary patterns, industry demand, work mode preferences, and geographical distribution efficiently.

This project showcases strong analytical thinking, dashboard design, and Excel-based business intelligence skills, making it suitable for data analytics portfolio presentation and entry-level Data Analyst opportunities.